The Affective Impact of **Prequestioning:** Reducing Test Anxiety Across Three Experiments

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Background

Test anxiety is regularly associated with poor learning outcomes, stress, and procrastination[1]. As students continuously face high-stakes exams, developing learning strategies that simultaneously promote learning and reduce test anxiety are essential.

Prequestioning (guessing before learning) improves learning outcomes[2], but its effects on test anxiety remain underexplored.

Across three experiments, we investigated whether a specific low-stakes implementation of prequestioning can reduce test anxiety.

Procedure

Stress Induction[3]
- Five challenging 'high-stakes' trivia questions

Anxiety Questionnaire
- State Trait Anxiety Inventory (STAI)[4]
- Performance Test Anxiety (PTA)
- Test Anxiety Inventory (TAI)[5]

Learning Intervention

Experiment 1 (*N* = 199, NUS)
- Prequestioning (PQ)
- Learning Objectives (LO)

Experiment 2 (*N* = 203, NUS)
- PQ
- No Activity (NA)

Experiment 3 (*N* = 152, Prolific)
- PQ
- LO
- NA

Anxiety Questionnaire

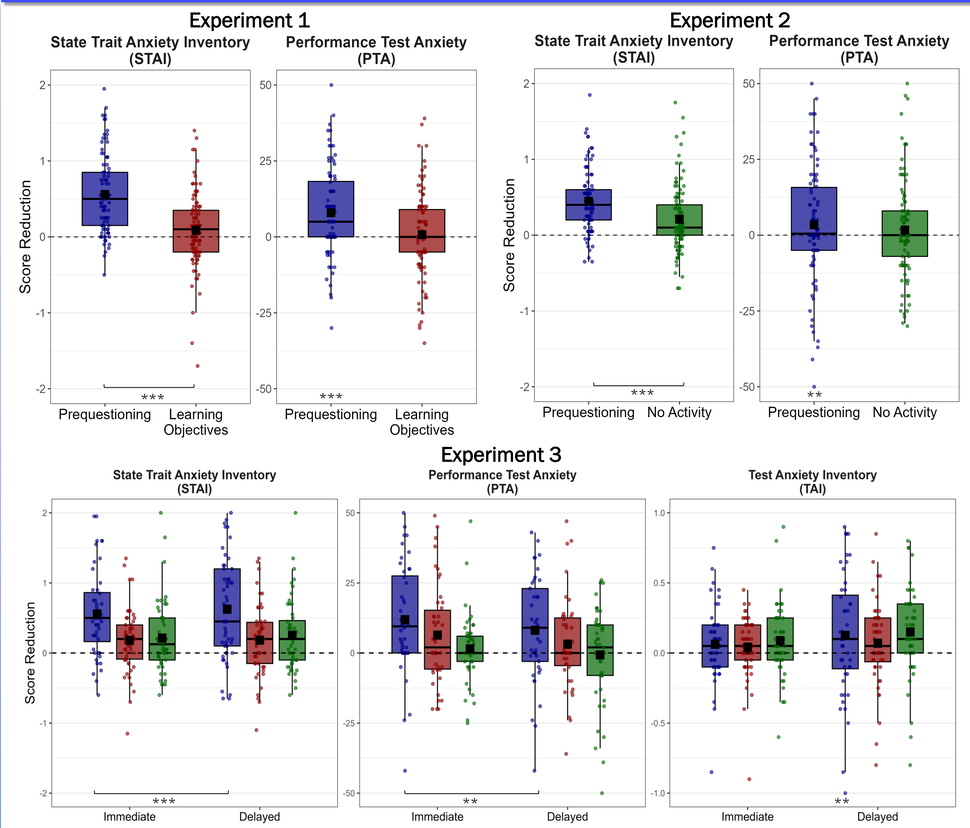
Immediate (E1 and E2) / 24-48H delay (E3)

Stress Reminders

Anxiety Questionnaire

Passage and Final Test

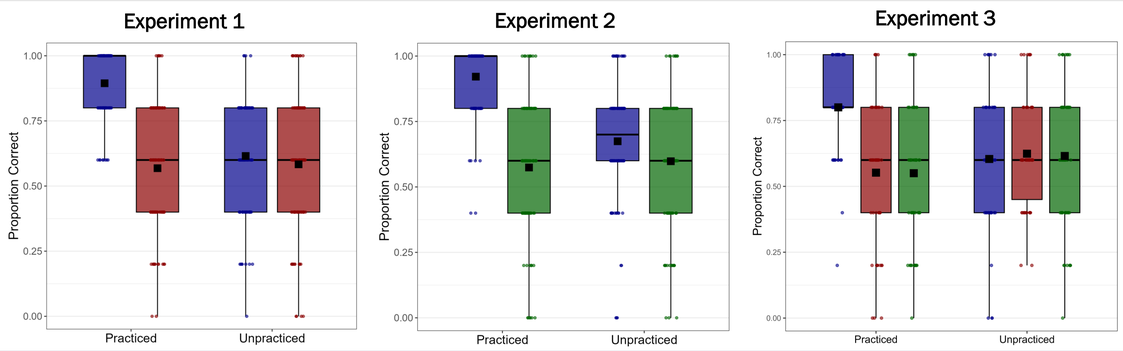
Test Anxiety Score Reduction by Measurement Type



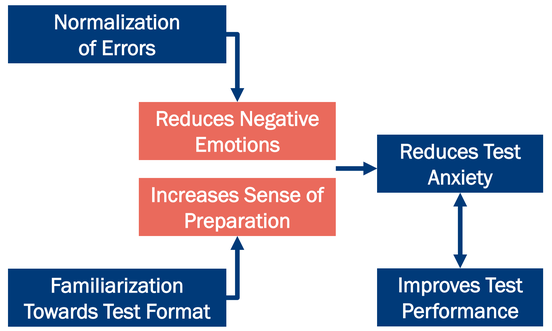
Prequestioning reported significantly greater immediate test anxiety reductions on the STAI and PTA compared to Learning Objectives (E1) or No Activity (E2).

E3: Prequestioning replicated immediate reductions in test anxiety on the STAI and PTA; test anxiety remained low at the 24-48H delay.

Criterial Test Performance by Question Type



Proposed Mechanisms



Implications and Next Steps

These experiments provide preliminary evidence that **prequestioning significantly reduces test anxiety compared to learning objectives or no activity, and that these anxiety reductions last over a 24-48H delay.**

Future research will examine prequestioning in classrooms and whether different instructional framing (e.g., positive vs neutral instructions) can moderate prequestioning's effectiveness in reducing test anxiety.

Educators may utilize prequestioning not only for its learning benefits, but also to potentially reduce test-related anxieties.

Further Information

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